MANGALORE UNIVERSITY

State Education Policy – 2024
[SEP-2024]

Question Bank

FOR

II SEMESTER BCA-GENERAL

BACHELOR OF COMPUTER APPLICATIONS

BCACACS202 -Data Structures using C(SEP)

UNIT – I

Questions carrying 2 marks:

1. Define data structure. Give two examples.
2. Write the classification of data structure.
3. Define primitive data structure. Give an example.
4. Define non-primitive data structure. Give an example.
5. Differentiate primitive data structure and non-primitive data structure.
6. Define linear data structure. Give an example.
7. Define non-linear data structure. Give an example.
8. Differentiate linear and non-linear data structure.
9. Name any 4 operations performed by data structure.
10. Define traversing and searching.
11. Define sorting. Why it is necessary
12. Define merging. Why it is necessary
13. Write any two algorithmic notations with examples.
14. Name any 2 types of control structures used in algorithms.
15. What is a sequence logic in algorithms? Give an example.
16. What is a iteration logic in algorithms? Give an example.
17. Write the structure of single alternative.
18. Write the structure of double alternative.
19. Write the structure of multiple alternatives.
20. Define Recursion
21. Define linear array. Write the formula to obtain number of data elements in an array.
22. Consider the linear array LA(1:10). Find the number of elements in an array.
23. Consider the linear array BB(-5:10). Find the number of elements in an array.
24. Consider the linear array AA(1:10). Suppose Base(AA)=100 and w=4 words per memory cell for AA. Find the address of AA[5].
25. Consider the linear array BB(5:50). Suppose Base(BB)=200 and w=4 words per memory cell for BB. Find the address of BB[15],
26. Consider the linear array CC(5:50). Suppose Base(CC)=300 and w=4 words per memory cell for CC. Find the address of CC[35]
27. Consider the two dimensional array X(5,5) . Suppose Base(X) = 100 and w=2 words per memory cell . Find the address of X[2,3] using row major order.
28. Consider the two dimensional array X(5,5) . Suppose Base(X) = 100 and w=2 words per memory cell . Find the address of X[2,3] using column major order.
29. Suppose a 5-element array A contains the values {25, 32, 65, 43, 78}. Find the values of the given arrays AB[2] and AB[4] after the loop.
    Repeat for K = 1 to 4:
    Set AB[K + 1] := A[K]
    [End of loop]
30. What is sparse matrix? Give example.
31. What is lower-triangular matrix? Give an example.
32. What is upper-triangular matrix? Give an example.
33. What is tri-diagonal matrix? Give an example.

Long Answer Questions

1) Write the classification of data structure and explain. (6)
2) List and explain the various operations performed on data structure. (6)
3) Explain the various algorithmic notation. (6)
4) Explain selection and iteration logic in detail with example. (6)
5) What do you mean by traversing a linear array? Write the algorithm to traverse a linear array. (5)
6) Explain the memory representation of one dimensional array (5)
7) Explain the representation of two-dimensional array in memory (5)
8) Write an algorithm to inserting an element into a linear array. (4)
9) Write an algorithm to delete an element from a linear array. (4)
10) Explain sparse matrix with example. (3)
11) Write an algorithm for TOWER OF HANOI using recursion with example (5)
12) Trace the following numbers using bubble sort for the following data:
    65, 25, 10, 80, 25, 11, 45,20 (5)
13) Trace the following list of numbers using insertion sort technique:
    23, 45, 56 22, 10, 27, 33, 48, 39,55,88,90 (5)
14) Trace the following list of numbers using selection sort technique:
    77, 33, 44, 11, 88, 22, 66, 55 (5)
15) Trace the following numbers using merge sort for the following data:
    65, 25, 10, 80, 25, 11, 45,20 (5)
16) Trace the following list of numbers using Quick sort technique:
    77, 33, 44, 11, 88, 22,66 ,55 (6)
17) Explain bubble sort with an example. (6)
18) Explain insertion sort with an example. (6)
19) Explain Selection sort with an example. (6)
20) Explain Merge sort with an example. (6)
21) Explain quick sort with an example. (6)
22) Write an algorithm for Bubble sort. (4)
23) Write an algorithm for Selection sort. (4)
24) Write an algorithm for Insertion sort. (4)
25) Write an algorithm for quick sort (5)

UNIT - II

Questions carrying 2 marks:

1. Define searching. Name 2 types of searching techniques.
2. What do you mean by linear/sequential search? Write its disadvantages.
3. What is the difference between linear and binary search?
4. Write one advantage and disadvantage of binary search.
5. What do you mean by dynamic memory allocation? Write any 2 functions used in ‘C’ for dynamic memory allocation.
6. What is linkedlist? List its different types.
7. What is free storage list?
8. What is garbage collection?

9. What role does the AVAIL list play in a linkedlist?
10. How is singly linked list terminated? Give diagram.
11. Define circularly linked list. Give its diagram.
12. What is two-way list? Give its diagram.
13. What is doubly linked list? Give its diagram.
14. Write the node diagram of two-way list.
15. What is underflow and overflow with respect to linked list?
16. Write the advantages of linked list over arrays.

Long Answer Questions

1) What is linear search? Write the algorithm for linear search. (5)
2) Explain binary search with an example. (6)
3) Write an algorithm for binary search. (5)
4) Explain any '2' dynamic memory allocation functions used in 'C' with syntax and example. (4)
5) Write a note on memory allocation and de-allocation function (4)
6) Write a note on dynamic memory management (4)
7) What is linked list? Explain the different types. (6)
8) Explain the memory representation of linked list. (5)
9) Write an algorithm for traversing a linked list. (4)
10) Write an algorithm for searching an item in a linked list. (4)
11) Write an algorithm to insert an item at the beginning of a linked list. (4)
12) Write an algorithm to insert an item after a given node of a linked list. (4)
13) Write an algorithm to insert an item at the end of a linked list. (4)
14) Write an algorithm to delete the first node from the linked list. (4)
15) Write an algorithm to delete a node following a given node of a linked list. (4)
16) Write a note on two-way lists. (4)
17) Write a note on header linked list (4)
18) Explain Circular linked list. (5)

UNIT - III

Questions carrying 2 marks:

1) What is stack? Name the basic operations performed on stack.
2) What is LIFO list? Mention any 2 its applications.
3) Mention any two applications of stack.
4) What do you mean by PUSH and POP operation?
5) Expand LIFO and FIFO.
6) What is a queue? Why queue is called FIFO list?
7) How does Stack differ from Queue?
8) Write an algorithm for PUSH operation using arrays.
9) Write an algorithm for POP operation using arrays.
10) Write an algorithm to find factorial of a number N using recursion.
11) Write an algorithm to generate Fibonacci series using recursion.
12) Define i) infix notation ii) prefix notation iii) postfix notation.

13) What do you mean by infix notation? Give an example.
14) What do you mean by prefix notation? Give an example.
15) What do you mean by postfix notation? Give an example.
16) What do you mean by polish and reverse polish notation?
17) Define queue. Name the primary queue operations.
18) Differentiate queue and circular queue.
19) What is circular queue? Give its diagram.
20) Define dequeue. What are its types?
21) What do you mean by input-restricted deque and output-restricted deque.
22) What is meant by priority queue? What is its use.

Long Answer Questions

1) Write an algorithm for PUSH and POP operations using arrays. (5)
2) Write an algorithm to push and POP operation onto a stack using linked list. (6)
3) Convert the given infix expression to postfix form. (any 3) (6)
   a) A * B + (C ^ D / E / F)
   b) a / b * c – d + e / f * (g + h)
   c) (X + Y / Z * W ^ P) –R
   d) (X + Y * Z ^ A / 4) – P
   e) (A + B) * (C – D) / E * F
   f) a * b + (c / d ^ e) - f
   g) (A - B * C ^ D) / (E + F)
   h) (a / b) * (c * f + (a – d) * e)
   i) A + B * (C – D * (E + F))
   j) a * b – c ^ d + e /f
4) Write an algorithm to evaluate postfix expression using stack (4)
5) Write an algorithm to convert infix to postfix expression using stack. (6)
6) Convert the following infix expression into postfix expression using stack status.
   Q: (A + (B * C – (D / E ^ F) * G) *H) (5)
7) Convert the following infix expression into postfix expression using stack status.
   Q: (((A + B) * D) ^ (E – F)) (5)
8) Evaluate the following postfix expression showing the stack status.
   P: 3, 5, +, 6, 4, -, *, 4, 1, -, 2, ^, + (5)
9) Evaluate the following postfix expression showing the stack status.
   P: 3, 1, +, 2, ^, 7, 4, -, 2, *, +, 5, - (5)
10) Evaluate the following postfix expression showing the stack status.
   P: 5, 6, 2, +, *, 1, 2, 4, /, ^ (5)
11) What is a Queue? Explain the different types of queues. (6)
12) Write an algorithm to perform insertion operation on queue using arrays. (4)
13) Write an algorithm to perform deletion operation on queue using arrays. (4)
14) Write an algorithm to perform insertion operation on queue using linked list. (4)
15) Write an algorithm to perform deletion operation on queue using linked list (4)
16) Write an algorithm to perform insertion operation on circular queue using array (4)
17) Write an algorithm to perform deletion operation on circular queue using array (4)
18) Write a note on i) circular queue. ii)Deque. (4)
19) What are Priority queues? What are the types? Explain (4)

UNIT-IV

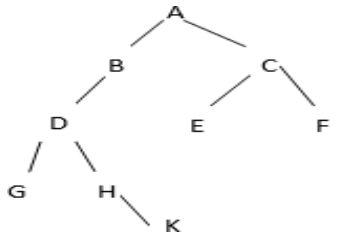
Questions carrying 2 marks:

1) Define binary tree. Give an example.
2) Define the terms with respect to tree.
   i) parent ii) left child iii) right child iv) siblings v) descendent vi) ancestor vii) edge viii) path ix) leaf node x) terminal node xi) branch xii) depth of a tree
3) Define complete binary tree. Give an example.
4) What is extended binary tree? Give an example.
5) What do you mean by internal nodes and external nodes?
6) If a node N occupies TREE [4], then in which the locations left child and right child are stored?
7) What is a binary search tree? Give an example.
8) Give any two applications of binary tree.
9) What is Graph? Give an example.
10) Define the terms (any two) with respect to graph.
    i) adjacent node ii) directed edge iii) undirected edge iv) directed graph v) undirected graph
11) Define adjacency matrix and path matrix

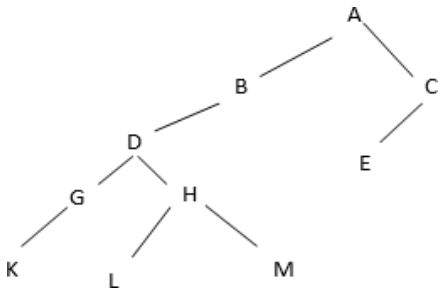
Long Answer Questions

1) Define the terms with respect to tree. (1 marks for each)
   i) parent ii) left child iii) rightchild iv) siblings v) descendent vi) ancestor vii) edge viii) path ix) leaf node x) terminal node xi) branch xii) depth of a tree (5)

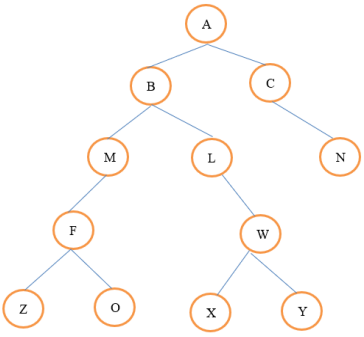
2) Write a note on i) Complete binary tree ii) Extended binary tree (4)
3) With an example, explain sequential representation of binary tree. (6)
4) What are the 3 standard ways of traversing a tree T with root R. write steps of each traversal using recursion. (6)
5) Draw the binary tree for the following (6)
   INORDER: E, A, C, K, F, H, D, B, G
   PREORDER: F, A, E, K, C, D, H, G, B
6) Draw the binary tree for the following inorder and postorder traversal. (6)
   INORDER: D, B, E, A, F, C, H,G
   POSTORDER: D, E, B, F H, G, C, A
7) Write the preorder, inorder and postorder traversal for the given tree. (6)



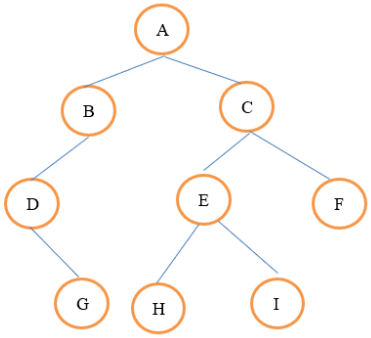
8) Write the preorder, inorder and postorder traversal for the given tree. (6)



9) Write the preorder ,inorder and postorder traversal for the given tree. (6)



10) Write the preorder, inorder and postorder traversal for the given tree. (6)



11) Draw the binary search tree for the following data and traverse in all 3 methods(Any one) (7)

i) 14, 15, 4, 9, 7, 18, 40, 35, 16, 13 ii)66, 26, 22, 34, 47, 79, 48, 32, 78

iii)G , C , R , T , A , D , X , L ,P , E

12) Draw the binary tree for the given algebraic expression: a *b– c ^ d + e / f Also write preorder, inorder and postorder traversal methods for the tree. (6)

13) Draw the binary tree for the expressions given below . Also traverse in preorder, post order and inorder. i) A * B + (C ^ D / E / F)

ii) a / b * c – d + e / f * (g + h) iii) (X + Y / Z * W ^ P) –R (7)

14) Explain the method of representing the graphs using sequential method with an example. (6)

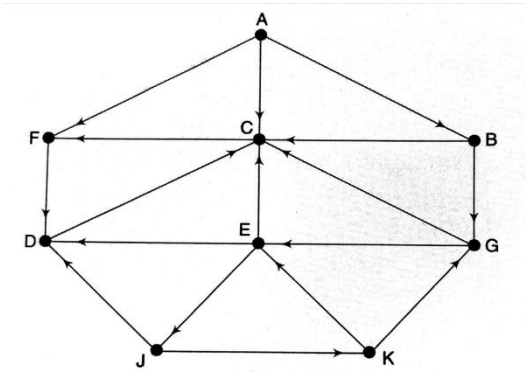
15) Explain Linked Representation of a graph with example (5)

16) What is adjacency matrix and path matrix, explain with an example. (6)

17) Write an algorithm for breadth first search (BFS) for a graph. (6)

18) Write an algorithm for depth first search (DFS) for a graph. (5)

19) Trace the graph above to reach J from node A using DFS algorithm. (7)



Object Oriented Programming using Java(SEP) (Theory)

Question Bank

UNIT I

Questions carrying 2 marks

1. What is Object Oriented Programming?
2. Define the terms i)Data Encapsulation ii)Data Abstraction
3. What do you mean by data hiding?
4. What is Polymorphism? Give example
5. What is dynamic binding?
6. Mention any four advantages of OOP.
7. What is Java Bytecode?
8. What is Java Virtual Machine?
9. Write the meaning of public static void main(String args[])
10. Mention two ways of writing comments in Java
11. List any 4 primitive data types available in Java.
12. What is the difference between10.3and10.3f?
13. What are the two ways of initializing variables?
14. What is dynamic initialization of variables? Give example.
15. List different arithmetic operators available in Java
16. What is type conversion? What are its types?
17. What is automatic(implicit)type conversion? Give example
18. Write the meaning of System.out.println().
19. Write a single print statement to produce the following output
    a. ONE
    b. TWO
    c. THREE
20. What is stream? List different types of streams in Java.
21. Differentiate byte and character streams.
22. List the predefined streams of system object
23. Write the syntax of simple if statement with example.
24. Give the syntax of declaring a symbolic constant with example.
25. List any 4 built in mathematical functions available in Java.

Questions carrying 4 or more marks

1. Explain various features of OOP.
2. Write a note on application and benefits of OOP.
3. List and explain different features of Java
4. Explain a simple java program with example.
5. Explain the java program structure with example.
6. What are identifiers? Write the rules to be followed when creating an identifier.
7. List and explain different primitive data types available in Java.
8. Explain automatic type conversion with suitable code example.
9. Explain explicit type conversion with example.
10. What are streams? List and explain different byte and character streams
11. Explain the relational and logical operators with example.
12. Explain the increment, decrement and conditional operator with syntax and example.
13. Explain the process of reading a character from the keyboard with suitable example.
14. Explain different forms of if statement with syntax and example
15. Explain switch statement with syntax and example

UNIT II

Questions carrying 2 marks

1. What is the purpose of break and continue statement? Give example.
2. Write the for statement for a loop that counts from 1000 to 0 by-2
3. Write a for loop to display even numbers between 1 and 100
4. What are arrays?
5. How to create and instantiate a one dimensional array? Give example
6. How to initialize an array at the time of creation? Give example.
7. What is the use of length member of arrays?
8. How to create and instantiate a two dimensional array? Give example
9. What are different ways of constructing String? Give example.
10. List any four methods associated with Strings.
11. What is vector? How it is different from array?
12. What are classes and objects?
13. What is the purpose of new operator?
14. Write the syntax to create a class in Java.
15. Write a statement that creates and instantiates an object in Java.
16. What are constructors? What is the purpose of a constructor?
17. What is the use of this keyword in java?
18. List any two restrictions of static methods.

Questions carrying 4 or more marks

1. Explain for loop with syntax and example.
2. Explain while loop with syntax and example
3. Explain do while loop with syntax and example.
4. Explain break and continue statement with suitable example.
5. Write a note on Labelled loops.
6. Explain the use of for each style of loop with suitable code example.
7. What are arrays? Explain how to declare instantiate initialize and use a one-dimensional array with suitable code example.
8. Explain different methods to initialize an array with suitable example.
9. Explain any four string methods with suitable example.
10. Explain any five vector methods with code example.
11. Explain the wrapper class methods String( ), valueOf( ) and parsing methods with syntax and example.
12. Write the general form of a class? Explain how to define a class in Java with suitable example.
13. Explain how to create an object in Java with suitable example.
14. Explain the use of Constructors with suitable example.
15. Explain command line arguments in Java with example.
16. Explain how to pass objects to methods with suitable example.
17. Explain how to return an object from a method with suitable example
18. Explain method overloading with suitable example.
19. What is recursion? Illustrate recursion with a code example.
20. Explain static variable and methods with suitable example.
21. Create a class Student with members Register number, name and marks in three subjects. Use constructors to initialize objects and write a method that displays the information of a student.
22. Explain constructor overloading with suitable example.
23. Explain how to pass variable-length arguments with suitable example.

UNIT III

Questions carrying 2 marks

1. What is inheritance?
2. What is the purpose of keyword extends?
3. What is the purpose of super( )?
4. What are abstract classes?
5. What are abstract methods?
6. How to prevent methods from overriding?
7. What are final variables?
8. What is the purpose of final class?
9. What are final methods?
10. How to define a package?
11. How to import a package?
12. List any four API packages of Java.
13. What are different methods of creating threads?
14. List different states of a thread.
15. List any four thread methods.
16. What is the purpose of notify( ) and notifyall( ) method?
17. Differentiate suspend( ) and stop( ) methods of a thread.
18. What is the purpose of resume() method?
19. What is the purpose of wait( ) and notify( )methods?
20. What is the purpose of exception handler?
21. List any four types of exceptions in Java.
22. What is the purpose of try and catch block?

Questions carrying 4 or more marks

1. Explain single inheritance with suitable example.
2. Explain method overriding with suitable example.
3. Explain multi-level inheritance with suitable example.
4. Explain the purpose of abstract and final classes.
5. Explain how to create and use a package in Java with suitable example.
6. List and explain different API packages available in Java.
7. Explain how to create and implement interface using suitable example.
8. Explain the process of creating a thread by Extended Thread class.
9. Explain the process of creating a thread by implementing Runnable interface with suitable example.
10. List and explain any five thread methods
11. Explain the life cycle of a Thread with diagram.
12. Explain exception handling in Java with suitable example.
13. Illustrate the use of multiple catch statements with suitable example.
14. Illustrate the use of finally with suitable example.
15. List and explain any 5 built-in exceptions of Java

Unit IV

Questions carrying 2 marks

1. What is an Applet?
2. List two types of Applets available in Java.
3. What is the purpose of appletviewer?

4. Write HTML file to run an applet program named Simple.
5. List any two features of Applets.
6. What is the purpose of paint( ) method?
7. List any four methods associated with Applets
8. What is the purpose of repaint( ) method?
9. Differentiate stop( ) and destroy( )methods.
10. What is the purpose of update( )method.
11. List the super classes of Applet defined by AWT
12. List any 4 methods defined by Applet.
13. What is an event? Give example.
14. What is the purpose of EventListener?
15. List the key features of a swing GUI
16. What happens when the constructor JButton(String msg) is called?
17. What is the purpose of setText( ) and getText( ) methods?
18. List any four swing components.
19. Differentiate Component and Container.

Questions carrying 4 or more marks

1. Write the complete Applet skeleton with example.
2. List and explain any 6 methods defined by Applet
3. List and explain the components of Delegation Event Model
4. List and explain any three EventListener Interfaces
5. Explain how any two Mouse events are handled in Applets with suitable example
6. Explain how any two Keyevents are handled in Applets with suitable example
7. With example explain creating frame window in applet.
8. Explain the purpose of JButton and explain any four methods associated with it.
9. Explain the use of JTextField and any four methods associate with it.
10. Explain the use of JList and any for methods associated with it.
11. Illustrate the use of JLabel with suitable example.
12. List and explain different methods associated with JCheckBox.
13. Explain the use of JCombobox and any for methods associated with it.
14. Explain different methods associate with JRadioButton control with suitable example.
15. Explain with example creating a Frame window in Applet.
16. Write a note on Swing Components and Containers.
17. Write and explain a simple swing application program.

II Semester BCA – Computational Mathematics

Question Bank

Unit-I

Part A -2 marks questions

1. What are the two types of errors?
2. Define Absolute, Relative and Percentage Error.
3. An approximate value of π is given by X_1=3.1428571 and its true value is X=3.1415926. Find the absolute and relative errors.
4. Find the relative error of the number 8.6 if both of its digits are correct.
5. Write the formula of Newton-Raphson Method.
6. Define Interpolation and Extrapolation.
7. If Δ is the forward difference operator then $\Delta^3 Y_3$=?
8. If ∇ is backward difference operator then $\nabla^3 Y_3$=?
9. If Y_1-Y_0=$\delta Y_{1/2}$ then $\delta^3 Y_{7/2}$.
10. Write Newton's forward difference interpolation formula.
11. Write Newton's backward difference interpolation formula.
12. Write Newton's General interpolation formula.

Part B-(4, 5, 6, 8 and 10 marks questions)

1. Find the real root of the following equation: $f(x)=x^3-x-1=0$ using Bisection method.
2. Find the real root of the following equation: $f(x)=x^3-2x-5=0$ using Bisection method.
3. Find the root of the equation 2x=cosx+3, correct to three decimal places using Iteration method.
4. Use the method of Iteration to find a positive root, between 0 and 1 of the equation $xe^x=1$.
5. Find the real root of the equation: $f(x)=x^3-2x-5=0$ using Method of False Position.
6. Use the Newton-Raphson method to find a root of the equation $x^3-2x-5=0$.
7. Find a root of the equation x sin x + cos x=0 using Newton-Raphson method.
8. Find the cubic polynomial which takes the following values y(0)=1, y(1)=0, y(2)=1 and y(3)=10. And obtain y(4) using Newton's forward difference interpolation formula.

9. The population of a town in the decennial census was as given below. Estimate the population for the year 1895 using Newton's forward difference interpolation formula.

Year(x)	1891	1901	1911	1921	1931
Population (y) (in thousands)	46	66	81	93	101

10. The population of a town in the decennial census was as given below. Estimate the population for the year 1925 using Newton's backward difference interpolation formula.

Year(x)	1891	1901	1911	1921	1931
Population (y) (in thousands)	46	66	81	93	101

11. In the table below the values of y are consecutive terms of a series of which the number 21.6 is the 6th term. Find the first term of the series using Newton's forward difference interpolation formula.

X	3	4	5	6	7	8	9
Y	2.7	6.4	12.5	21.6	34.3	51.2	72.9

12. In the table below the values of y are consecutive terms of a series of which the number 21.6 is the 6th term. Find the tenth term of the series using Newton's backward difference interpolation formula.

X	3	4	5	6	7	8	9
Y	2.7	6.4	12.5	21.6	34.3	51.2	72.9

13. The table below gives the values of tan x for $0.10 \leq X \leq 0.30$. Find tan 0.40 using Newton's backward difference interpolation formula.

X	0.10	0.15	0.20	0.25	0.30
Y=tan x	0.1003	0.1511	0.2027	0.2553	0.3093

14. From the following table, find the number of students who obtained less than 45 marks.

marks	30-40	40-50	50-60	60-70	70-80
No.of students	31	42	51	35	31

15. From the following table of values of x and f(x), determine f(0.23).

X	0.20	0.22	0.24	0.26	0.28	0.30
f(x)	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139

16. Find the 7th term and the general term of the series 3,9,20,38,65,…

17. Certain corresponding values of x and $\log_{10}x$ are (300, 2.4771), (304, 2.4829), (305, 2.4843) and (307, 2.4871). Find $\log_{10}$ 301 using Lagrange's Interpolation Formula.

18. If $y_1=4$, $y_3=12$, $y_4=19$, $y_x=7$, find x using Lagrange's interpolation formula.

19. Find the Lagrange's interpolating polynomial of degree 2 approximating the function y=ln x defined by the following table of values. Hence determine the value of ln 2.7.

X	2	2.5	3.0
Y=ln x	0.69315	0.91629	1.09861

20. The function y=sin x is tabulated below

X	0	$\pi/4$	$\pi/2$
Y=sin x	0	0.70711	1.0

Using Lagrange's interpolation formula find the value of sin $(\pi/6$.

21. Given the table of values

X	150	152	154	156
Y-$\sqrt{x}$	12.247	12.329	12.410	12.490

Evaluate $\sqrt{155}$ using Lagrange's interpolation formula.

22. Certain corresponding values of x and $\log_{10}$x are (300, 2.4771), (304, 2.4829), (305, 2.4843) and (307, 2.4871). Find $\log_{10}$ 301 using Newton's divided difference Formula.

23. Using the following table find f(x) as a polynomial in x using Newton's general interpolation formula.

X	-1	0	3	6	7
f(x)	3	-6	39	822	1611

UNIT-II

Part A -2 marks questions

1. Write the equation for fitting a straight line for $\partial S/\partial a_0$and$\partial S/\partial a_1$.
2. What is a power function?
3. What do you mean by a polynomial of nth degree?
4. What is Exponential function?
5. Write the formula of curve fitting by sum of exponentials.
6. Write Newton's forward difference formula for $\left[\frac{dy}{dx}\right]x = x0$and $\left[\frac{d2y}{dx2}\right]x = x0$
7. Write Newton's backward difference formula for $\left[\frac{dy}{dx}\right]x = x0$and $\left[\frac{d2y}{dx2}\right]x = x0$
8. Write the general formula for Numerical Integration.
9. Write Trapezoidal rule and Simpson's 1/3 rule.
10. Write the Simpson's 3/8 rule for $\int_{x0}^{xn} ydx$.

Part B-(4, 5, 6, 8 and 10 marks questions)

1. The table below gives the temperature T (in ^oC) and lengths l (in mm) of a heated rod. If $l=a_0+a_1T$, find the best values for a_0 and a_1.

T	20°	30°	40°	50°	60°	70°
l	800.3	800.4	800.6	800.7	800.9	801.0

2. Certain experimental values of x and y are given below

x	0	2	5	7
y	-1	5	12	20

If $y=a_0+a_1x$, find approximate values of a_0 and a_1.

3. Fit the polynomial of a second degree to the data points given in the following table.

x	0	1.0	2.0
y	1.0	6.0	17.0

4. Determine the constants a and b by the method of least squares such that $y=ae^{bx}$, fit the following the data

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

5. From the following table of values of x and y, obtain dy/dx and d^2y/dx^2 for x=1.2 using Newton's forward difference formula.

X	1.0	1.2	1.4	1.6	1.8	2.0	2.2
Y	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

6. Calculate the first and second derivatives of the function tabulated below at the point x=2.2 using Newton's backward difference formula.

X	1.0	1.2	1.4	1.6	1.8	2.0	2.2
Y	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

7. From the following table of values of x and y, obtain dy/dx and d^2y/dx^2 for x=1.6 using Newton's forward difference formula.

X	1.0	1.2	1.4	1.6	1.8	2.0	2.2
Y	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

8. From the following table of values of x and y, obtain $d(J_0)/dx$ and at x=0.1 using Newton's forward difference formula.

X	0.0	0.1	0.2	0.3	0.4
$J_0(x)$	1.0000	0.9975	0.9900	0.9776	0.9604

9. From the following table of values of x and y, obtain dy/dx and when x=6 using Newton's forward difference formula.

X	4.5	5.0	5.5	6.0	6.5	7.0	7.5

Y	9.69	12.90	16.71	21.18	26.37	32.34	39.15

10. From the following table of values of x and y, obtain dy/dx and d^2y/dx^2 for x=3 using Newton's forward difference formula.

X	0	1	2	3	4	5	6
Y	6.9897	7.4036	7.7815	8.1291	8.4510	8.7506	9.0309

11. From the following table of values of x and y, obtain dy/dx and d^2y/dx^2 for x=6 using Newton's backward difference formula.

X	0	1	2	3	4	5	6
Y	6.9897	7.4036	7.7815	8.1291	8.4510	8.7506	9.0309

12. Find the Area bounded by the curve and the x-axis from x=7.47 to x=7.52 from the following table using Trapezoidal rule.

X	7.47	7.48	7.49	7.50	7.51	7.52
f(x)	1.93	1.95	1.98	2.01	2.03	2.06

13. Evaluate I= $\int_0^1 1/(1+x)\,dx$ by trapezoidal rule when h=0.25, correct to three decimal places.
14. Evaluate I= $\int_0^1 1/(1+x)\,dx$ by trapezoidal rule and Simpson's 1/3 rule when h=0.5, correct to three decimal places.
15. Evaluate I= $\int_0^1 \sqrt{1-x^2}$dx by trapezoidal rule and Simpson's 1/3 rule when h=0.2.
16. Evaluate I=$\int_0^1 \sin \pi x dx$ using Simpson's 1/3 rule when h=0.5.
17. Evaluate I=$\int_1^3 \frac{1}{x} dx$ by Simpson's 1/3 rule with 4 strips.
18. Evaluate I=$\int_1^3 \frac{1}{x} dx$ by Simpson's 1/3 rule with 8 strips.
19. Evaluate I=$\int_0^\pi tsin\ tdt$ using Trapezoidal rule when h=$\pi/4$.
20. Evaluate I=$\int_{-2}^2 tdt/(5+2t)$ using Trapezoidal rule when h=1.
21. Find the value of $\int_3^7 x^2$logx dx by taking 4 strips using Simpson's 1/3 rule.
22. Find the approximate value of $\int_0^{\pi/2} \sqrt{\cos\theta}\, d\theta$ by dividing the interval into six parts using Simpson's 3/8 rule.
23. Evaluate I=$\int_0^1 \cos x\, dx$using Simpson's 3/8 rule when h=0.2.

UNIT-III

Part A -2 marks questions

1. Define square matrix. Give example.
2. Define lower triangular and upper triangular matrix. Give example.
3. Define symmetric and skew-symmetric matrix. Give example.
4. If A=$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and B= $\begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}$ find (AB)'

5. If A=$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ show that AI=A.
6. Show that A=$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is Orthogonal.
7. Show that the matrix B=$\begin{bmatrix} -2/3 & 1/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \\ 1/3 & -2/3 & 2/3 \end{bmatrix}$ is orthogonal.
8. If A=$\begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{bmatrix}$ show that A'= -A.
9. Find the inverse of the matrix A=$\begin{bmatrix} 5 & -2 & 4 \\ -2 & 1 & 1 \\ 4 & 1 & 0 \end{bmatrix}$
10. Given the matrix A=$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ find $\|A\|_1$, $\|A\|_2$ and $\|A\|_\infty$
11. For the matrix A=$\begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ compute A^2 and A^4.
12. Compute the Inverse of the matrix A=$\begin{bmatrix} 2 & 4 & 3 \\ 0 & 1 & 1 \\ 2 & 2 & -1 \end{bmatrix}$
13. If A=$\begin{bmatrix} 2 & 3 & 2 \\ 1 & 0 & 0 \\ 2 & 0 & 2 \end{bmatrix}$ and B=$\begin{bmatrix} -1 & 2 & 1 \\ 2 & 3 & 4 \\ 1 & -2 & 3 \end{bmatrix}$ form AB and BA.

Part B-(4, 5, 6, 8 and 10 marks questions)

1. Express the matrix A=$\begin{bmatrix} 1 & 7 & 8 \\ 6 & 2 & 9 \\ 5 & 4 & 3 \end{bmatrix}$ as a sum of symmetric and a skew-symmetric matrix.
2. Solve the following equations using matrix inversion method.
   3x+y+2z=3
   2x-3y-z=-3
   x+2y+z=4
3. Solve the following system using Gauss Elimination method.
   2x+y+z=10
   3x+2y+3z=18
   x+4y+9z=16
4. Solve the following system using Gauss Elimination method.
   5x-2y+z=4
   7x+y-5z=8
   3x+7y+4z=10
5. Solve the following system using Gauss Jordan method.
   2x+y+z=10
   3x+2y+3z=18

$x+4y+9z=16$

6. Solve the following system using Gauss Jordan method.

   $5x-2y+z=4$
   $7x+y-5z=8$
   $3x+7y+4z=10$

7. Solve the equations using LU Decomposition method.

   $2x+3y+z=9$
   $x+2y+3z=6$
   $3x+y+2z=8$

8. Solve the following system using LU Decomposition method.

   $5x-2y+z=4$
   $7x+y-5z=8$
   $3x+7y+4z=10$

9. Use Jacobi's Method to solve following system of linear equations.. Carry out 3 iterations.

   $10x_1-2x_2-x_3-x_4=3$
   $-2x_1+10x_2-x_3-x_4=15$
   $-x_1-x_2+10x_3-2x_4=27$
   $-x_1-x_2-2x_3+10x_4=-9$

10. Use Gauss Seidal Method to solve the following system. Carry out 3 iterations.

    $10x_1-2x_2-x_3-x_4=3$
    $-2x_1+10x_2-x_3-x_4=15$
    $-x_1-x_2+10x_3-2x_4=27$
    $-x_1-x_2-2x_3+10x_4=-9$

11. Solve the following system using Jacobi's method.Carry out 3 iterations.

    $10x+2y+z=9$
    $2x+20y-2z=-44$
    $-2x+3y+10z=22$

12. Solve the following system using Gauss-Seidal method.Carry out 3 iterations.

    $10x+2y+z=9$
    $2x+20y-2z=-44$
    $-2x+3y+10z=22$

13. Find the solution, to three decimals of the system using Jacobi's method.

    $83x+11y-4z=95$
    $7x+52y+13z=104$
    $3x+8y+29z=71$

14. Find the solution, to three decimals of the system using Gauss-Seidal method.

    $83x+11y-4z=95$
    $7x+52y+13z=104$
    $3x+8y+29z=71$

UNIT-IV

Part A -2 marks questions

1. Write the Taylor's series for y(x).
2. Write Euler's general formula.
3. Given y'= - y, y(0)=1, h=0.01. Find y(0.01) using Euler's method.
4. Write Runge-Kutta 4th order formula.
5. Write Adams-Moulton predictor-corrector formula.
6. Write Milne's Predictor formula.
7. Write Milne's Corrector formula.

Part B-(4, 5, 6, 8 and 10 marks questions)

1. From the Taylor's series for y(x), find y(0.1) correct to four decimal places if y(x) satisfies $y'=x-y^2$ and y(0)=1.
2. Given dy/dx-1=xy and y(0)=1, obtain the Taylor's series for y(x) and compute y(0.1),correct to four decimal places.
3. Using Euler's method solve the differential equation (dy/dx)+2y=0, where h=0.1 and obtain y(0.1), y(0.2), y(0.3).
4. Using Euler's method solve the differential equation $(dy/dx)-1= y^2$, where h=0.1 and obtain y(0.1), y(0.2), y(0.3).
5. Solve by Euler's method, the equation dy/dx=x+y, y(0)=0. Choose h=0.2 and compute y(0.4) and y(0.6).
6. Determine the value of y when x=0.1 given that y(0)=1 and $y'=x^2+y$, h=0.05, using Euler's modified method.
7. Determine the value of y(2) in steps of 0.1, given that y(1)=1 and $(dy/dx)-\sqrt{xy}=2$, h=0.05, using modified Euler's method.
8. Given that y(0)=1 and $(dy/dx)=x^2+y$, determine y(0.02), y(0.04) and y(0.06) using modified Euler's method.
9. Given dy/dx=y-x where y(0)=2, and h=0.1. Find y(0.1) using Runge-Kutta fourth order formula.
10. Given dy/dx=y-x where y(0)=2, and h=0.1. Find y(0.2) using Runge-Kutta fourth order formula.
11. Given $dy/dx=1+y^2$ where y=0 when x=0, and h=0.2. Find y(0.2) using Runge-Kutta fourth order formula.
12. Given $dy/dx=1+y^2$ where y=0 when x=0, and h=0.2. Find y(0.6) using Runge-Kutta fourth order formula.
13. Use the Runge-Kutta fourth order method to find the value of y when x=1 given that y=1 when x=0 and dy/dx=y-x/y+x.
14. Use the Runge-Kutta method to solve $10(dy/dx)=x^2+y^2$, y(0)=1 for the range $0\leq x \leq 0.4$ with h=0.1.
15. Using Adams-Moulton formula, solve $y'=1 + y^2$ where y=0 when x=0, and h=0.2. Find y(0.8). Given that y(0.2)=0.2027, y(0.4)=0.4228, y(0.6)=0.6841.

16. Use Adams-Moulton formula for tabulating a solution of $10(dy/dx)=x^2+y^2$, $y(0)=1$ for the range $0.5 \leq x \leq 1.0$
17. Tabulate the solution of $dy/dx=x+y$, $y(0)=0$ for $0.4 \leq x \leq 1.0$ with $h=0.1$ using Adams-Moulton formula.
18. Use Milne's formula for tabulating a solution of $10(dy/dx)=x^2+y^2$, $y(0)=1$ for the range $0.5 \leq x \leq 1.0$
19. Tabulate the solution of $dy/dx=x+y$, $y(0)=0$ for $0.4 \leq x \leq 1.0$ with $h=0.1$ using Milne's formula.
20. Using Milne's formula solve $y'=1+y^2$ where $y(0)=0$, and $h=0.2$. Find $y(0.8)$. Given that $y(0.2)=0.2027$, $y(0.4)=0.4228$, $y(0.6)=0.6841$.
21. Using Milne's formula solve $y'=1+y^2$ where $y(0)=0$, and $h=0.2$. Find $y(1.0)$. Given that $y(0.2)=0.2027$, $y(0.4)=0.4228$, $y(0.6)=0.6841$.
22. A boundary value problem is defined by $y''+y+1=0$, with boundary conditions $y(0)=0$, $y(1)=0$ with $h=1/4$. Determine the value of $y(0.5)$ using finite difference method.
23. Solve the boundary value problem $d^2y/dx^2=y$ with boundary conditions $y(0)=0$, $y(2)=3.627$ with $h=0.5$ by using finite-difference method.

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